PROJECT PLAN / PROPOSAL

SavePlate – Smart Food Waste Reduction App

Task 1: Project Plan / Proposal

Group Name	Clovers
Team Leader	Nikhil Rana
Team Members	1. Nikhil Rana (Team Leader) 2. Aditya Kumar Jaiswal 3. Semon Chakhradhar
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Lecturer	Seetha Letchumi
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Table of Contents

Contents

1. Overview

1.1 Management Summary

SavePlate is a smart food waste reduction application that we will design it to address the growing food waste problem in Malaysian households. We will draw inspiration from the documented issue of excessive food waste in Malaysia, the SavePlate app aims to empower users to take control of their food inventory, reduction in unnecessary waste and contribution to more sustainable community through intelligent meal planning and donation facility.

Our application will be developed as a mobile platform which includes six core use cases which include user registration with privacy settings, food inventory management, food browsing and donation, food analytics and reporting, notification and weekly meal planning. Our project is undertaken by our Group Clovers, a team of three students from the BIT216 Software Engineering Principles course at Mid Valley International Collage affiliated with HELP University.

1.2 Motivation

Food waste is a serious environmental and economic issue in whole world including country like Malaysia. Many households often discard food simply due to poor inventory management, expiry date, and lack of planning. Our app SavePlate is conceived as a practical solution to address these problems at the household level. By providing real time inventory tracking, expiry date alerts and meal planning features, our app will enable users to make smarter decisions about their food consumptions, and this will help to reduce waste and encourage people to handle food more properly and more responsibly.

1.3 Customer / Target Users

The primary users of SavePlate are primarily targeted to Malaysian household members who wish to manage their food inventory more efficiently, reduce waste, and engage with community donation system and our Secondary users include community volunteers and neighbors who benefit from the donation listings by browsing available food items that are near to them.

1.4 What the Project Will Deliver

SavePlate will deliver a fully functional mobile application with these following points:

- User registration and privacy/security management.
- Our food inventory management system will track expiry products.
- We will also add food donation and functions like module browsing.
- It will also have visual analytics and dashboard.
- Our system will also provide automated notification for expiry, donation and meal reminders.
- Weekly meal planner on available inventory.

1.5 New Product or Extension

SavePlate will be a new product. While there are also other existing food waste reduction app globally, Our SavePlate will be primary target for Malaysian household context, incorporation donation workflows, dual language consideration and a hyper local community feature that is distinguish from the existing solutions.

1.6 Cost and Duration

Item	Details
Estimated Development Cost	RM 0 (It is an academic project with 0 commercial cost)
Tools & Platforms	Open-Source APIs, Vs Code, Android Studio, GitHub, Gantt Project and Figma.
Project Duration	14 weeks (4 June 2026 – 3 September 2026)
Team Size	3 members (Group: Clovers)

2. Project Aims

Our project aims of the SavePlate project are intentional statements of our team that hopes to achieve upon project completions:

- To design and develop a user-friendly Universal Interface, technology driven application that assists Malaysian household in reducing food waste through intelligent inventory tracking that will help in households.
- To communicate facility in community engagement by enabling food donation between households and promoting a culture of sharing social responsibility.
- To empower users with data driven insights about their food saving behavior through a proper explained visual analytics and reports.
- To provide a reliable meal planning tool that will help to maximize the user's available food inventory and help users to plan meals around what they have.
- At the end, we ensure that user data security and privacy through robust and strict authentication mechanisms including 2FA and MFA.

3. Project Objectives

The following measurable objectives define the specific outcomes the project will accomplish:

1. **Deliver a working user registration and privacy module:** Use Case 1 with 2FA support will end by 28 July 2026.
2. **Implement a food inventory management system:** Use Case 2 that allows users to log, edit and track food items with expiry dates and other functions like storage categories.
3. **Develop a browse and donation interface:** Use Case 3 enables users to search, filter, and claim the donated food in the community.
4. **Build a food analytics and reporting dashboard:** Use Case 4 that displays visual progress like indicators and historical data on food saved and donations that are made.
5. **Implement an automated notification system:** Use Case 5 will help to alert users on expiry dates, donation status and meal planning reminders.
6. **Create a weekly meal planner:** Use Case 6 will link the inventory items to planned meal, and the schedules reminder will be based on the ingredient that is available.

7. **Complete and submit all project documentation:** As the academic requirement by the specified deadline will be done in time (Task 1: 12 June 2026).

4. Project Scope

4.1 In Scope

The following features and deliverables are within the scope of the SavePlate project:

- User account registration, login and privacy and security setting that include 2FA and MFA configurations. – Use Case 1
- Food inventory management like adding, editing, deleting and category food items with expiry tracking. – Use Case 2
- Food browsing and donation system like posting, browsing, filtering and claiming food donations. – User Case 3
- Food analytics like visual dashboard like displaying food saved, waste reduction and donations. – Use Case 4
- Notification system like automated alert for expiry, donation updated and meal reminders. – Use Case 5
- Weekly meal planners like planning meals for existing inventory with system suggested recipes. – Use Case 6
- Deployment in GitHub with a public repository that will be maintained by team members.
- Full project documentation like proposal, requirements specifications and design document.

4.2 Out of Scope

Here are the scopes for the SavePlate project:

- Integration of free open-source stores APIs.
- Monetization and Payment features.
- Use of Advance AI/ML recipe’s generations.
- Integration of Physical hardware.
- Multi-language support.
- Commercial deployment on android store (Play Store).